MULTI-SITE SOLAR PV PORTFOLIO (38.7 kW)

Date: 21-Sep-2026

Portfolio: Karnataka Tech Portfolio (2 Sites, 3 Buildings) | Location: Karnataka | Grid-Tied LT Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
38.71 kW	**61,376 kWh**	**Rs. 1,825,300**	**4Y 9M**
39.60 kWp Installed	CUF 18.1% | PR 75%	Rs. 18.25 Lakhs	Profit: Rs. 180.15L

[SITE] Customer & Site Engineering Parameters

Verified project configuration and sizing pathway

Company / Customer Name	Karnataka Tech Portfolio
Location / Address	Karnataka
Contact Email Address	vidya@voltroof.com
Design Path Selected	Rooftop Area Analysis
System Target Capacity	38.71 kW (2 Sites Portfolio)
Total Installed Array Rating	39.60 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	72 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	4975.00 Sq.Feet (462.19 sq.m)
Utilized Usable Rooftop Area	4214 Sq.Feet (391.49 sq.m)
First Year Annual Generation	61,376.24 kWh
25-Year Cumulative Generation	1,363,648.94 kWh
Estimated Payback Period	4 Years & 9 Months
Post-Payback Profitability (25 Yrs)	Rs. 18,015,437.81 (INR 180.15 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 8,811,999.18 (INR 88.12 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 1118 Tonnes CO2 | Tree Equivalence: 1788 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[PORTFOLIO] Multi-Site Portfolio Allocation & Regulatory Schedule

Aggregated LT solar portfolio overview, site-wise capacities, and statutory compliance

TOTAL SITES	TOTAL CAPACITY	TURNKEY CAPITAL	ANNUAL YIELD
PORTFOLIO	AGGREGATE	12-ITEM BOM	CLEAN POWER
2 Sites	**38.71 kW**	**Rs. 1,825,300**	**61,376 kWh**
3 Buildings Total	72 Panels (550W)	Rs. 18.25 Lakhs	1st Year Aggregate

[ALLOCATION] Site-by-Site Allocation & Engineering Capacity Matrix

Independent LT net-meter installations conforming to 20 kW per-connection statutory ceiling

Sl.	Site Name & Location	Bldgs	Gross Area	Capacity	Inverter	1st Yr Yield	Turnkey Cost
01	Vidyanagar Campus (Shivamo..	2 Nos.	2,775 sq.ft	20.00 kW *	20 kW 3-Ph	31,711 kWh	Rs. 940,300
02	Technology Park (Bengaluru..	1 Nos.	2,200 sq.ft	18.71 kW	19 kW 3-Ph	29,665 kWh	Rs. 885,000
	TOTAL PORTFOLIO SUMMARY		**4,975 sq.ft**	**38.71 kW**	**39 kW Total**	**61,376 kWh**	**Rs. 1,825,300**

** Marked installation(s) capped at statutory 20.0 kW Low Tension (LT) ceiling in accordance with KERC regulations.*

KERC & ESCOM LOW TENSION (LT) 20 kW REGULATORY COMPLIANCE MANDATE

Under Karnataka Electricity Regulatory Commission (KERC) Rooftop Solar PV regulations and ESCOM interconnect guidelines, individual Low Tension (LT) net-metered service connections are subject to a statutory capacity ceiling of 20.0 kW per consumer installation / meter account. Interconnections above 20 kW necessitate High Tension (HT 11 kV) infrastructure, dedicated transformer bays, and elevated statutory approvals.

To maintain full regulatory compliance while meeting multi-site commercial / institutional power demand, this portfolio is architected into discrete, independent LT installations. Each individual site operates with its own dedicated bidirectional net-meter, solar inverter, ACDB protection board, and safety switchgear, while engineering specifications, equipment warranties, and turnkey financials are consolidated within this unified dossier.

LT PORTFOLIO STATUTORY & GRID INTERCONNECTION STANDARDS

Metering Architecture:	Independent Bi-Directional TOD Net-Meters per site (1 - 20 kW LT Class)
Anti-Islanding Protection:	Mandatory automatic grid disconnection upon utility outage (IEC 62116 / IEEE 1547 compliant)
Switchgear & Isolation:	Dedicated IP65 ACDB with Type-II AC/DC SPDs & 10 kA Breaking MCBs per connection
DISCOM Permitting:	Separate online DISCOM portal filings with aggregated turnkey EPC execution and common monitoring

[ARRAY] Vidyanagar Campus -- Building 1: Layout & Array Sizing

Site: Vidyanagar Campus (Shivamogga, Karnataka) | Sub-Array: 13.44 kW | Site Inverter: 20 kW

Building / Array Identification	Vidyanagar Campus -- Admin Block (Rooftop Array)
Gross Rooftop Area	1575.00 Sq.Feet (146.32 sq.m)
Utilized Effective Rooftop Area	1344 Sq.Feet (124.86 sq.m)
Building Plant Target Capacity	13.44 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	25 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	20 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (20 kW)

Core inverter rating, MPPT topology, and operating parameters

Selected Inverter Rating	20 kW Grid-Tied Solar Inverter
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx35 Sq.mm
DC Array Isolator Switch	80A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	40 Amps
Rated Inverter Operating Current	27.82 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx35 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx16 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	25.0 Meters (ACDB to Utility Net-Meter)

[ARRAY] Vidyanagar Campus -- Building 2: Layout & Array Sizing

Site: Vidyanagar Campus (Shivamogga, Karnataka) | Sub-Array: 9.99 kW | Site Inverter: 20 kW

Building / Array Identification	Vidyanagar Campus -- Library Block (Rooftop Array)
Gross Rooftop Area	1200.00 Sq.Feet (111.48 sq.m)
Utilized Effective Rooftop Area	999 Sq.Feet (92.81 sq.m)
Building Plant Target Capacity	9.99 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	19 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	20 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (20 kW)

Core inverter rating, MPPT topology, and operating parameters

Selected Inverter Rating	20 kW Grid-Tied Solar Inverter
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx35 Sq.mm
DC Array Isolator Switch	80A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	40 Amps
Rated Inverter Operating Current	27.82 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx35 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx16 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	25.0 Meters (ACDB to Utility Net-Meter)

[ARRAY] Technology Park -- Building 1: Layout & Array Sizing

Site: Technology Park (Bengaluru, Karnataka) | Sub-Array: 18.71 kW | Site Inverter: 19 kW

Building / Array Identification	Technology Park -- Innovation Lab (Rooftop Array)
Gross Rooftop Area	2200.00 Sq.Feet (204.39 sq.m)
Utilized Effective Rooftop Area	1871 Sq.Feet (173.82 sq.m)
Building Plant Target Capacity	18.71 kW
Solar PV Module Rating	550 W Mono PERC / TOPCon Half-Cut
Number of Solar Modules	35 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	19 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (19 kW)

Core inverter rating, MPPT topology, and operating parameters

Selected Inverter Rating	19 kW Grid-Tied Solar Inverter
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx35 Sq.mm
DC Array Isolator Switch	80A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	40 Amps
Rated Inverter Operating Current	26.43 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx35 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx16 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	18.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV Modules (Mono PERC / TOPCon	550W Mono PERC Half-Cut Cells (>21.3% Efficiency)	72 Nos.	Rs. 720,000
02	On-Grid Solar String Inverter(s)	Dual MPPT, IP65 (Vidyanagar Campus: 20 kW, Technolog	2 Set(s)	Rs. 410,000
03	Rooftop mounting structure	HDGI, module-compatible	2 Site(s)	Rs. 190,000
04	DC solar cable	6 sq.mm copper, PV-rated	2 Site(s)	Rs. 35,700
05	AC cable	35 sq.mm copper	2 Site(s)	Rs. 84,500
06	MC4 connectors	PV-rated connector pairs	2 Site(s)	Rs. 21,600
07	DCDB	Isolator + fuse + Type-II DC SPD	2 Site(s)	Rs. 35,000
08	ACDB	40A MCB + 63A RCCB + Type-II AC SPD	2 Site(s)	Rs. 35,000
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	2 Site(s)	Rs. 16,500
10	Lightning arrester	Standard rooftop PV type	2 Site(s)	Rs. 15,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	2 Site(s)	Rs. 72,000
12	Installation & commissioning	Labour and commissioning	2 Site(s)	Rs. 190,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 1,825,300.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 2,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 4,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 23.6 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 10,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 10,000 (Refundable utility escrow)

STATUTORY COMPLIANCE & NET-METERING POLICY

All electrical installations conform to CEA (Technical Standards for Connectivity of Distributed Generation Resources) Regulations 2013 and State Grid Code. Net-metering bi-directional billing operates under applicable KERC / State SERC LT Solar Tariff orders.

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 1,825,300	**4Y 9M**	**Rs. 180.15L**	**Rs. 88.12L**
12-Item BOM Basis	Break-Even Year	Rs. 18,015,438	Alternative Asset

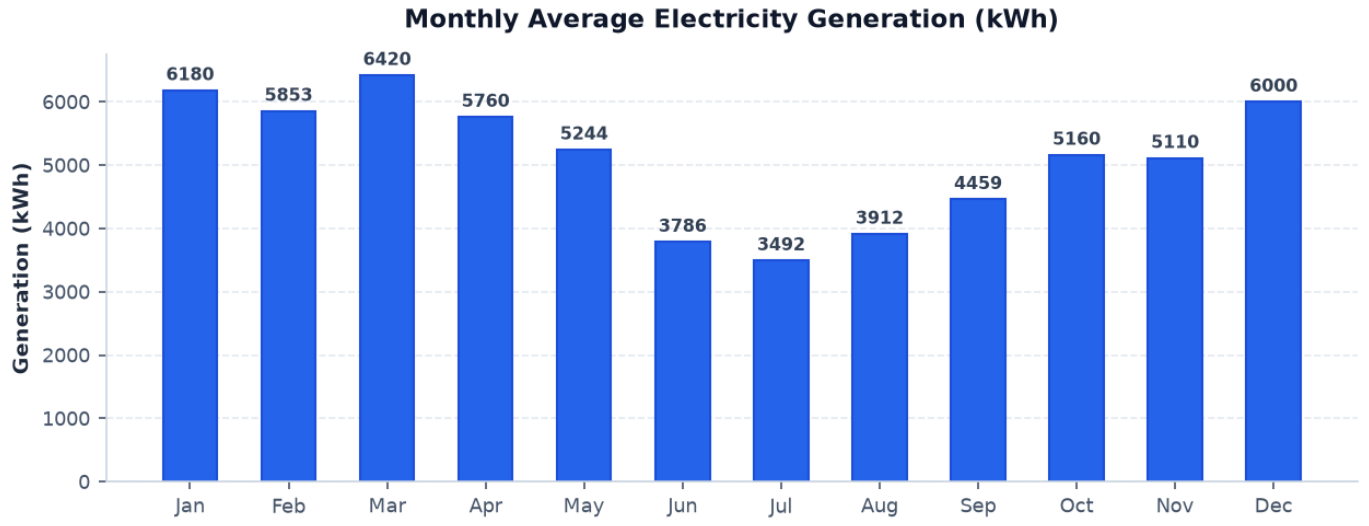
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	61,376	Rs. 8.05	494,079	23,226	470,853	470,853
02	60,762	Rs. 8.45	513,443	23,226	490,217	961,070
03	60,155	Rs. 8.87	533,574	23,226	510,348	1,471,417
04 *	**59,553**	**Rs. 9.32**	**555,037**	**23,226**	**531,811**	**2,003,228**
05	58,958	Rs. 9.78	576,607	23,226	553,381	2,556,609
06	58,368	Rs. 10.27	599,441	23,226	576,215	3,132,824
07	57,785	Rs. 10.78	622,917	23,226	599,691	3,732,515
08	57,207	Rs. 11.32	647,579	23,226	624,353	4,356,869
09	56,635	Rs. 11.89	673,385	23,226	650,159	5,007,028
10	56,068	Rs. 12.48	699,732	23,226	676,506	5,683,534
11	55,508	Rs. 13.11	727,704	23,226	704,478	6,388,012
12	54,952	Rs. 13.76	756,146	23,226	732,920	7,120,933
13	54,403	Rs. 14.45	786,123	23,226	762,897	7,883,830
14	53,859	Rs. 15.18	817,579	23,226	794,353	8,678,182
15	53,320	Rs. 15.93	849,393	23,226	826,167	9,504,350
16	52,787	Rs. 16.74	883,657	23,226	860,431	10,364,780
17	52,259	Rs. 17.57	918,195	23,226	894,969	11,259,750
18	51,737	Rs. 18.45	954,542	23,226	931,316	12,191,066
19	51,219	Rs. 19.37	992,118	23,226	968,892	13,159,958
20	50,707	Rs. 20.34	1,031,383	23,226	1,008,157	14,168,115
21	50,200	Rs. 21.36	1,072,273	23,226	1,049,047	15,217,162
22	49,698	Rs. 22.43	1,114,727	23,226	1,091,501	16,308,663
23	49,201	Rs. 23.55	1,158,685	23,226	1,135,459	17,444,122
24	48,709	Rs. 24.73	1,204,575	23,226	1,181,349	18,625,471
25	48,222	Rs. 25.96	1,251,842	23,226	1,228,616	19,854,088

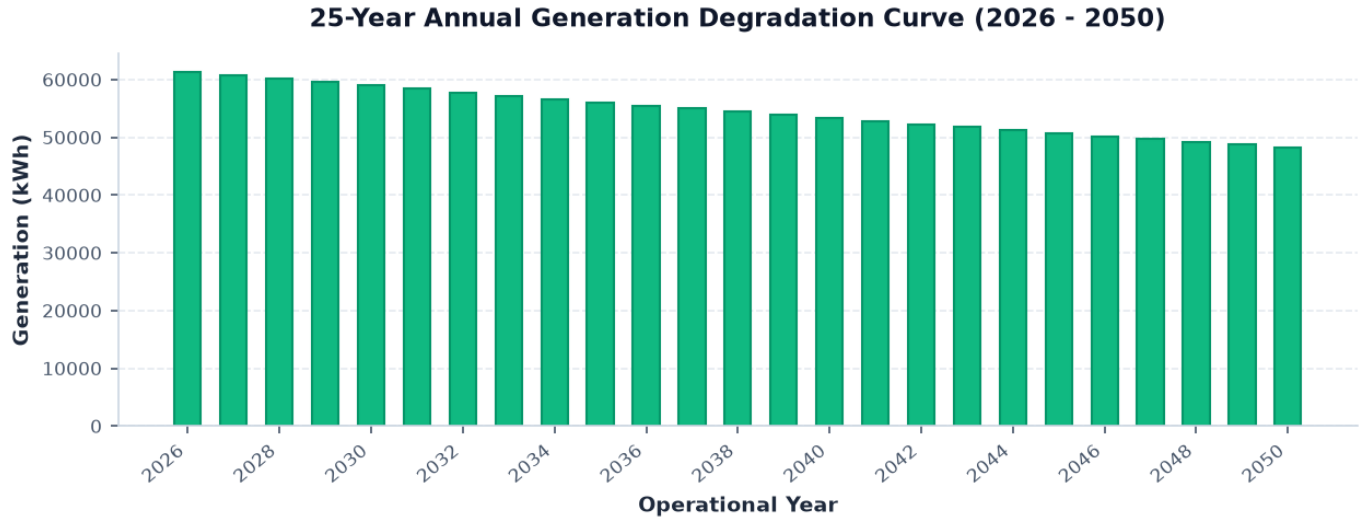
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1586 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

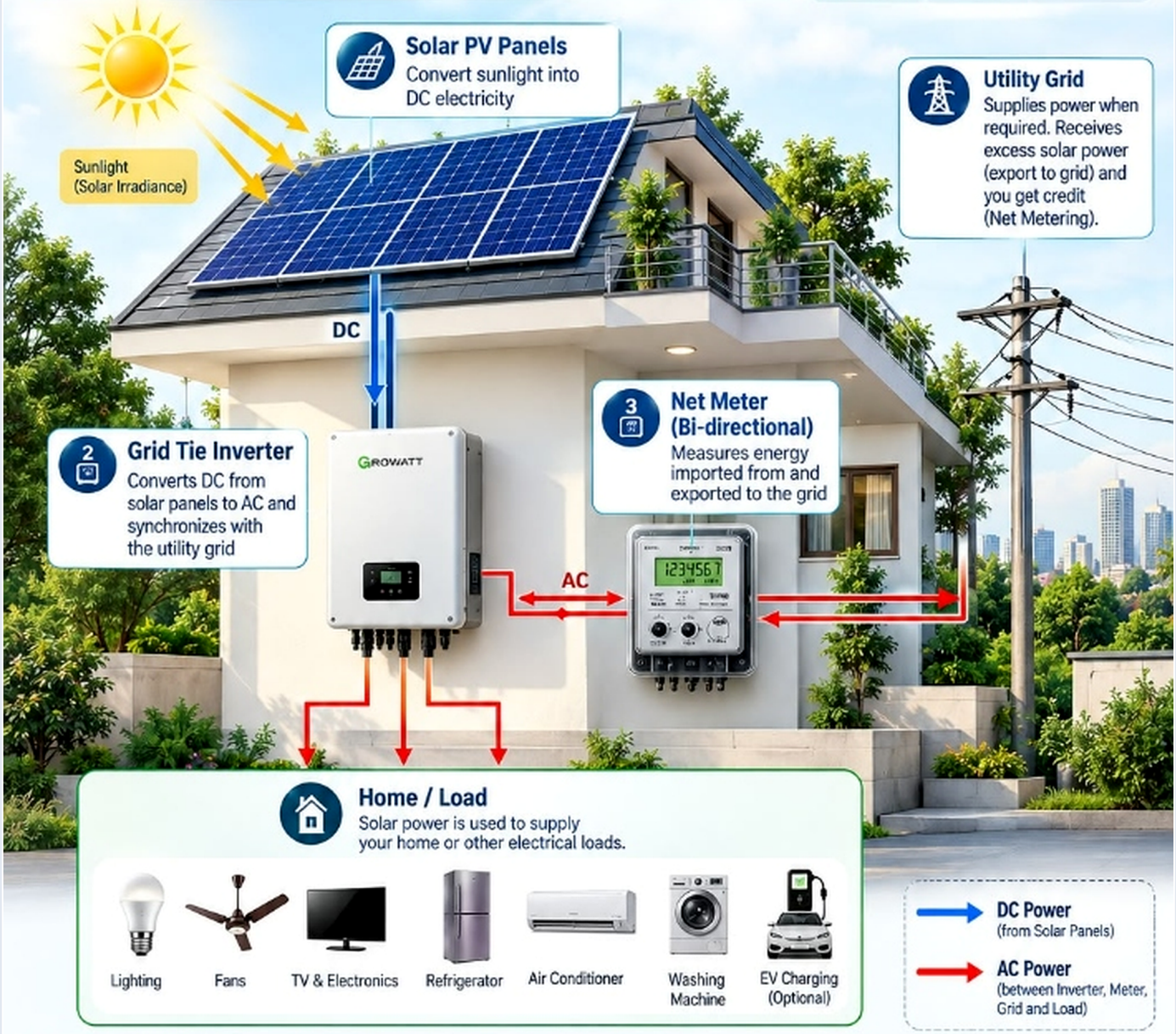
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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